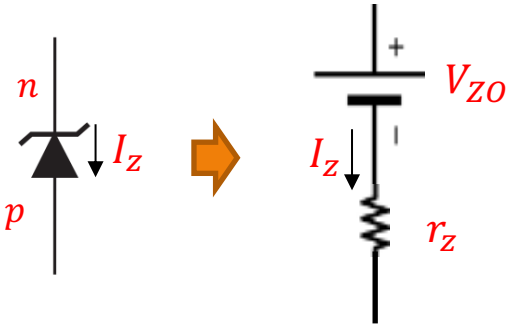
Diode Applications

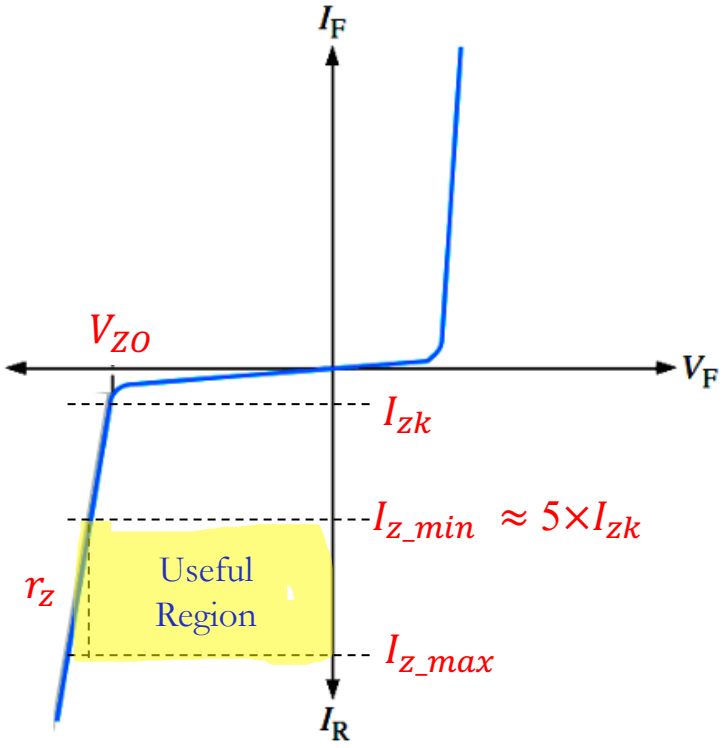
- Regulation Using Zener Diodes.
- Signal Shaping.
- Diode Types.

Zener Diode Model

The Zener diode is modeled using the constant battery-R model. Three parameters need to be known (datasheet): V_{ZO}, r_z and I_{zk}.

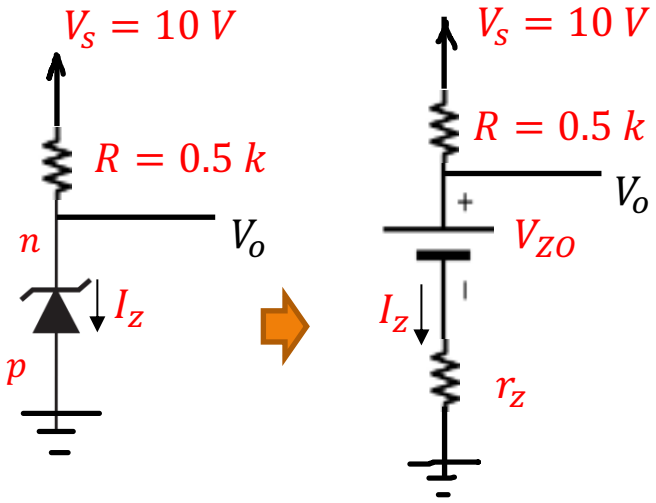


$$V_z = V_{ZO} + I_Z\, r_z$$



Example

1) The circuit below uses a Zener diode with the following parameters. The voltage source has no variations. $V_Z = 6.8\,V\ at\ 5\,mA, r_z = 20\,\Omega$ and $I_{zk} = 0.2\,mA$.



$$V_z = V_{ZO} + I_Z\, r_z \qquad V_{ZO} = V_Z - I_Z\, r_z$$

$$V_{ZO} = 6.8 - (5\,m)(20) = 6.7\,V$$

Is the diode in breakdown? YES

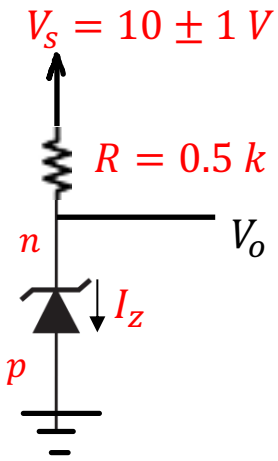
$$I_z = \frac{V_s - V_{ZO}}{R + r_z}$$

$$I_z = \frac{10 - 6.7}{500 + 20} = 6.35\,mA$$

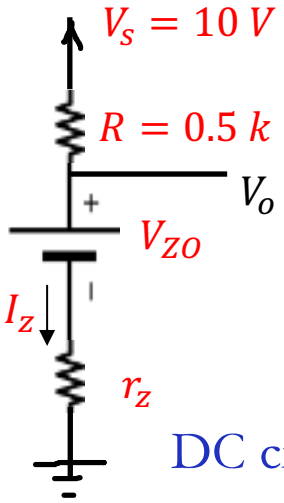
$$V_o = V_Z = V_{ZO} + I_Z\, r_z = 6.7 + (6.35\,m)(20) = 6.83\ \text{V}$$

Example

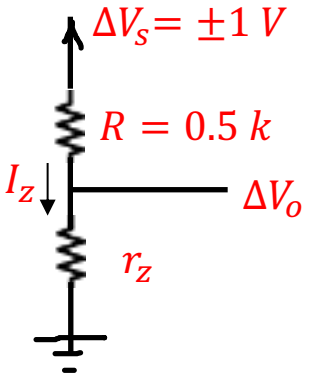
2) Now the source has a ±10% change. What will be the change in the output voltage?







DC circuit



AC circuit

±10% change in source

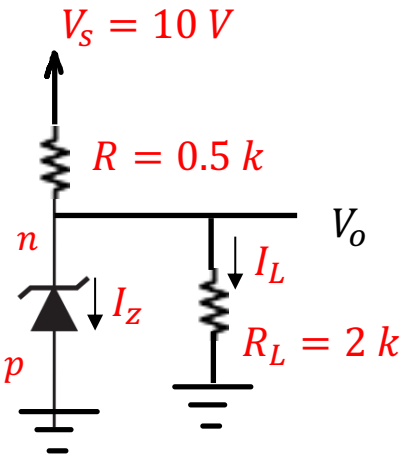
±0.6% change in output

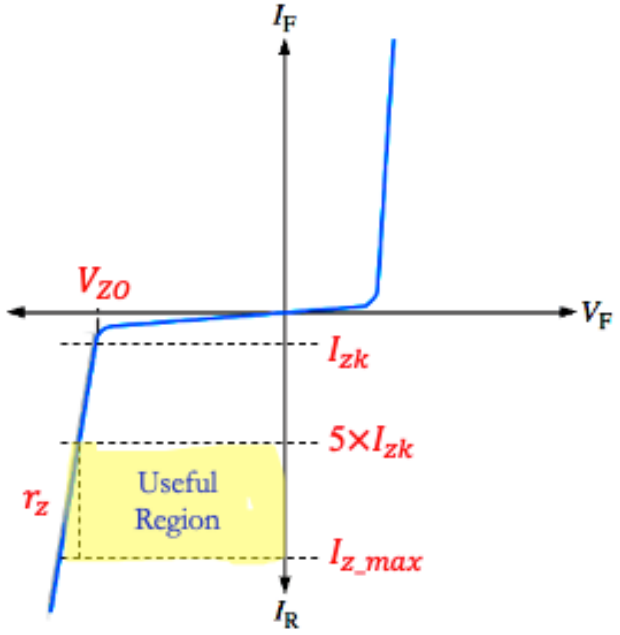
$$\Delta V_o = \Delta V_s \frac{r_z}{R + r_z}$$

$$\Delta V_o = \pm 1 \frac{20}{500 + 20} = \pm 38.5\ mV$$

Example

3) The source now is fixed at 10 V, but a load of 2 kΩ is connected. What is the change in output voltage?





$$I_L = \frac{V_o}{R_L} = \frac{6.83}{2\ k} = 3.42\ mA$$

$$I_z = 6.35 - 3.42 = 2.93\ mA$$

$$V_o = V_{ZO} + I_Z\, r_z = 6.7 + (2.93\ m)(20) = 6.76\ V$$

1% change in output

What if load is 0.5 k Ω?

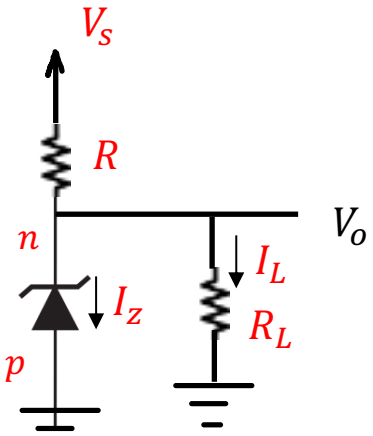
$$I_L = 6.83/2k = 13.66\ mA$$

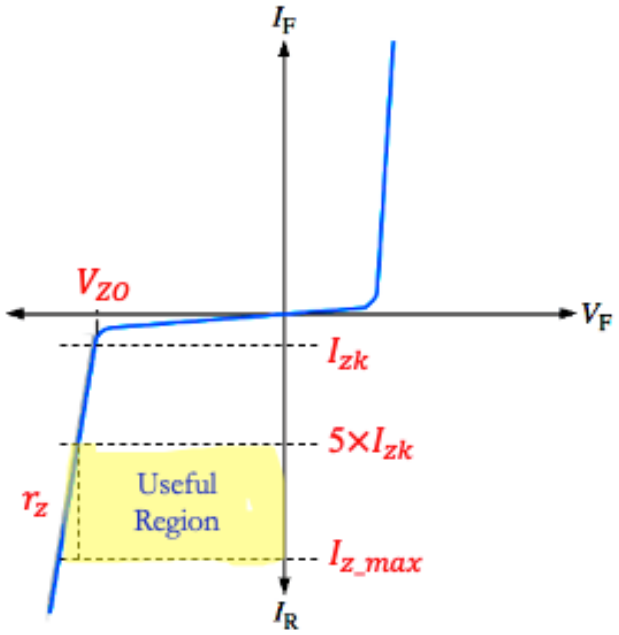
Diode is OFF

What is R_{L_min}?

Example

The resistance R should not be too high. Otherwise the diode will not get the minimum current for useful region?

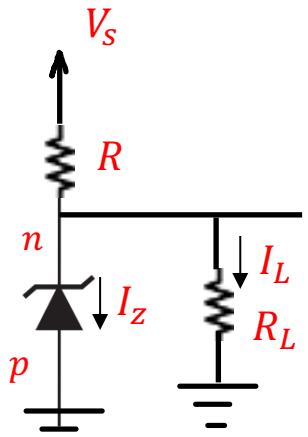




$$R_{max} = \frac{V_{s_min} - V_{ZO} - r_z\, I_{Z_min}}{I_{Z_min} + I_{L_max}}$$

Example

4) Design a shunt regulator to get an output voltage around 10 V. Use the 1N7470. The input source is expected to have 20±25% V and the load can take current up to 20 mA.

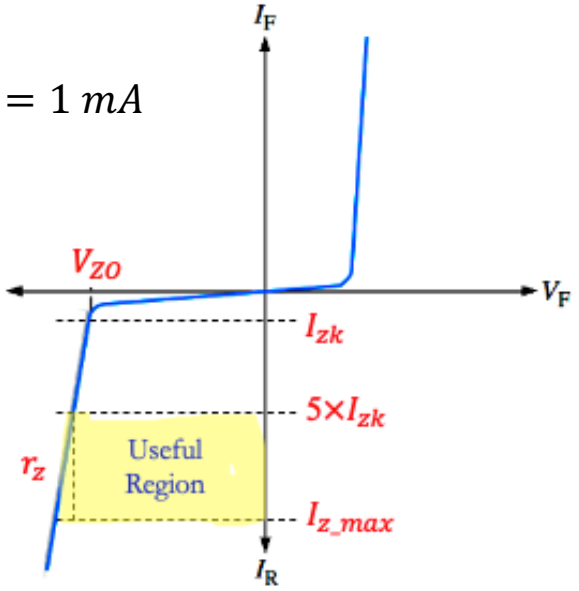


1N 7470: $V_Z = 10\ V\ at\ 25\ mA, r_z = 7\ \Omega$ and $I_{zk} = 1\ mA$

$$V_{ZO} = V_Z - I_Z\, r_z$$

$$V_{ZO} = 10 - (25\ m)(7) = 9.825\ V$$

$$R_{max} = \frac{V_{s_min} - V_{ZO} - r_z\, I_{Z_min}}{I_{Z_min} + I_{L_max}}$$



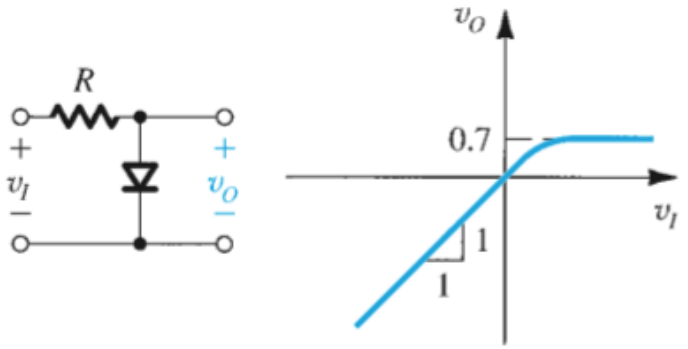
$$R_{max} = \frac{15 - 9.825 - (7)\ (5\ m)}{5\ m + 20\ m} = 205\ \Omega$$

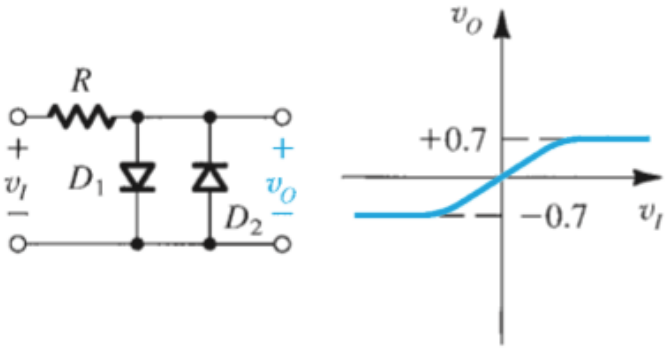
$$R \leq 205\ \Omega$$

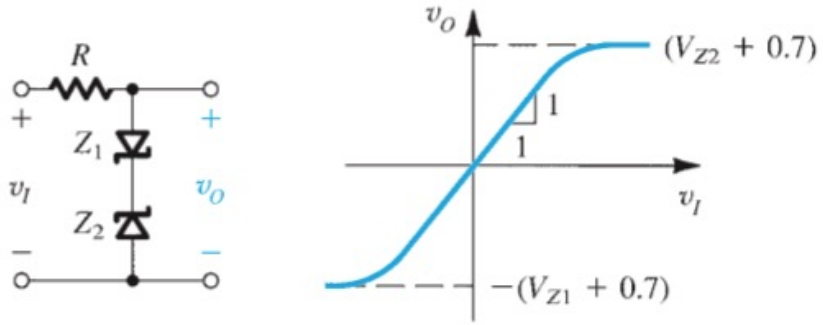
Other Diode Applications

Diode circuits work very well as waveform shaping circuits.

Clamping/clipping/limiting circuits



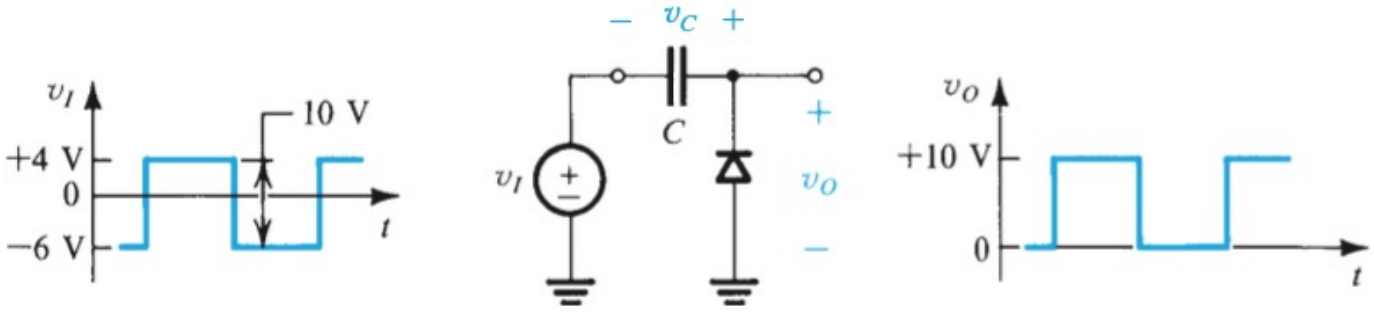


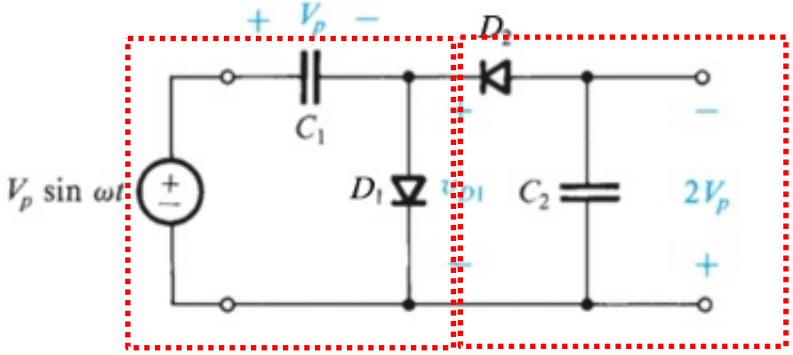


Other Diode Applications

Diode circuits work very well as waveform shaping circuits.

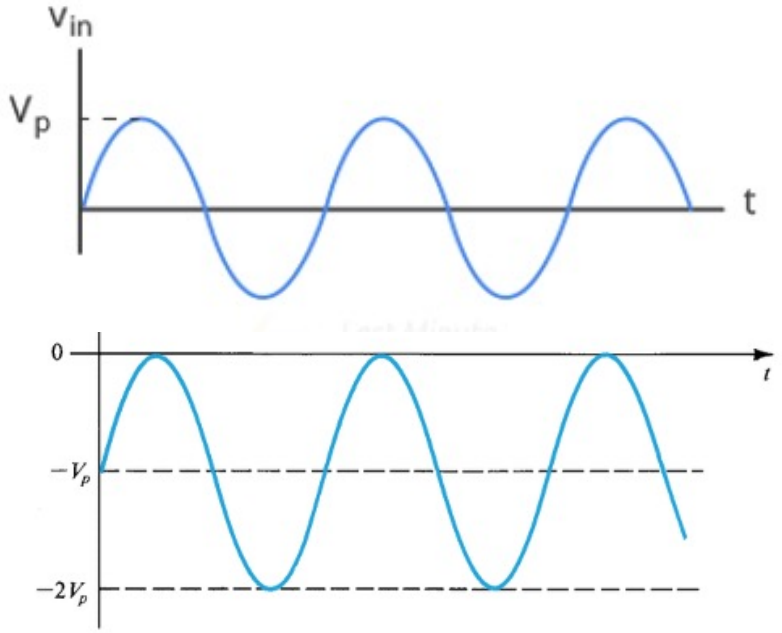
Shape forming circuits





Shift down the input signal

Peak detector



Diode Types

Regular (switching)

Zener (regulator)

Schottky diodes

Gunn/IMPATT/BARRITT

Tunneling diodes

Photodiodes/PIN/APD

Laser diodes/LED

Thyristors

Q & A

Send me an email if
you have any
questions.